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Thermal conductivities of silicon and germanium in solid and liquid states measured by non-stationary hot wire method with silica coated probe

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Abstract

The thermal conductivities of silicon and germanium have been determined using the non-stationary hot wire method. Measurements were carried out over the temperature range 293–1724 K on solid and liquid silicon and on liquid germanium in alumina tube. For solid silicon, the thermal conductivities were about 139 W/mK at 293 K and 19 W/mK at 1573 K and displayed temperature dependence steeper than T^{-1} , where T is the temperature. Calculation of thermal conductivities for solid silicon based upon isotope, three-phonon and four-phonon scatterings indicates that phonon conduction dominates heat conduction at temperatures below 1000 K. At temperatures above 1000 K, on the contrary, contributions from electron, hole and electron–hole pair to heat conduction became greater progressively with a temperature rise. For liquid silicon, the thermal conductivity was about 57 W/mK at 1700 K and exhibited a slight increase with an increase in temperature. The thermal conductivity of liquid germanium was about 43 W/mK at 1273 K and slightly increased with increasing temperature. In both liquids, temperature dependency of thermal conductivity values was discussed from the view point of the Wiedemann–Franz law. © 2002 Elsevier Science B.V. All rights reserved.

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1. Introduction

In recent years, the mathematical modeling of the heat and fluid flow in high temperature processes have been established as a useful method

for improving process control and product quality. In the on-going development of the Czochralski technique for single crystal growth, one of the primary requirements is for the production of silicon single crystal having higher quality as well as larger diameters. To meet this requirement, stricter process design and control should be carried out based upon the mathematical modeling for various steps in the crystal growth process. The modeling should come first for heat transfer across the interface between the crystal and the melt,

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because this principally determines the temperature distribution near the interface, which has a decisive effect on the quality of the product. To model the heat transfer, it is required to have accurate data for thermal conductivities of both solid and liquid silicon at high temperatures.

From the theoretical aspect, the thermal conductivities of solid and liquid semiconductors are expected to be fundamental data which would aid the investigation of the heat conduction mechanism and the electronic structure in semiconductors and the applicability of Wiedemann–Franz law to their liquids. With respect to the thermal conductivity for solid silicon, many data measured over wide temperature ranges were collated by Touloukian et al. [1]. Furthermore, Parrot and Stuckes suggested that phonons dominate heat conduction at lower temperatures and electrons and holes also contribute more as temperature rises [2]. In addition, Glassbrenner et al. [3] theoretically described the temperature dependence of phonon thermal conductivity, and Price [4] and Drabble et al. [5] that of electronic thermal conductivity. However, there is considerable discrepancy between the experimental [3,6] and theoretical values. To understand the heat conduction mechanism, it is still necessary to obtain accurate data for thermal conductivities of solid silicon.

There have been a few reports about thermal conductivities of liquid silicon. Yamamoto et al. [7] measured thermal diffusivities of liquid silicon at the melting point by the laser flash method and derived thermal conductivity values using the density and heat capacity. Kimura et al. [8] also obtained thermal conductivity values in the same way over the temperature range between the melting point and 1780 K. With respect to liquid germanium, Taylor et al. [9] and Takasuka et al. [10] measured thermal diffusivities at the temperature ranges between 1223 and 1330 K and between melting point and 1330 K, respectively, and derived thermal conductivity values. It should be noted that there have been no measurements of the thermal conductivity on liquid silicon and germanium. Because of a lack of precise values for the density and heat capacity of liquid silicon and germanium, the thermal conductivity derived from

the thermal diffusivity would be less reliable than that obtained without using other physical properties of the sample.

The authors have previously developed the non-stationary hot wire method with silica-coated probe for applying to liquid metals [11], by which the thermal conductivity can be directly measured without using other physical properties of the sample. In the previous work, the thermal conductivities for liquid mercury and lead could be successfully determined with no effects of coating when the thickness of the coating layer is between 50 and 200 μm . Furthermore, it has been concluded that this method can be applied to materials having higher melting point, as long as materials for probe coating are pertinent. Consequently, the aims of the present work are to accurately measure thermal conductivities of silicon and germanium in solid and liquid states over a wide temperature range using the non-stationary hot wire method and to discuss heat conduction mechanisms in these materials.

2. Experimental procedure

2.1. Sample

Samples of silicon were Fz (100)-plane single crystal rods (30 mm diameter and 40 mm length). The total concentration of impurities ranged between 2×10^{15} and $3 \times 10^{15} \text{ cm}^{-3}$. Prior to the measurements, the samples were cleaned with hydrofluoric acid solution and then rinsed in distilled water, in anhydrous ethanol and finally in acetone. On the other hand, samples of germanium were prepared from germanium ingot, the purity of which was 99.99% and the specific resistance was $0.47 \Omega\text{m}$ at room temperature.

2.2. Measurement

In the non-stationary hot wire method, electrical power is supplied to a thin metal wire (hot wire) placed in the sample, which serves as both a heating element and a temperature sensor, and the

temperature rise (ΔT) of the hot wire is recorded continuously. The thermal conductivity of the sample (λ) is obtained from Eq. (1) [12]

$$\Delta T = \frac{Q}{4\pi\lambda} (\ln t + A), \quad (1)$$

where Q is the heat generation rate per unit length of the wire, t the time and A a constant. In practice, the thermal conductivity of the sample is derived using the slope of the linear portion of the relation between ΔT and $\ln t$ from Eq. (2).

$$\lambda = \frac{Q}{4\pi} \left/ \frac{d\Delta T}{d\ln t} \right. \quad (2)$$

When measurements are carried out on electrically conducting samples such as liquid silicon, considerable difficulty is encountered since there is electrical leakage to the sample from the hot wire. However, this problem can be overcome by forming a silica insulator layer of less than 100 μm in thickness on the wire [11].

Fig. 1 shows schematic diagrams of the experimental apparatuses for solid silicon and liquid silicon and germanium used in the present work. In the measurements of solid silicon, the probe was closely sandwiched between two cylindrical silicon single crystals, the samples of which were covered with about 100 nm thick SiO_2 film formed by thermal oxidation. The probe consisted of a hot wire (0.15 mm diameter and 30–40 mm long) of platinum-13% rhodium to which two platinum potential leads (0.15 mm diameter) were attached 20–25 mm apart to allow four-terminal resistance measurements. These wires were connected to platinum lead wires of 0.5 mm diameter.

The construction of the probe for liquid silicon and germanium was basically the same as that for solid silicon. The lead and potential wires were supported by alumina tubing. To form a silica layer on the wires, the probe was dipped into silica slurry (Toh-a Gousei Kagaku: ARON CERAMIC CC) several times and then dried in ambient atmosphere for 24 h to remove moisture, followed by sintering at 423 K for 2 h. The thickness of the silica layer ranged between 80 and 100 μm . It has been confirmed in the previous work [11] that silica films with these thicknesses give no effects on thermal conductivity values. In detail, to consider

the effect of the coating layer, Eq. (1) can be modified as follows:

$$\Delta T = \frac{Q}{4\pi\lambda} \left[\ln t + A + \frac{1}{t} (B \ln t + C) \right], \quad (3)$$

where B and C are the constants which can be calculated from the physical property values (thermal conductivity, thermal diffusivity and so on) of the hot wire, the coating layer and sample. The comparison between Eqs. (1) and (3) indicates that the latter term in Eq. (3) indicates the effect of the coating layer. In the cases of liquid germanium at 1473 K, the values of A , B and C are 5.85×10^1 , -4.45×10^{-3} and -1.73×10^{-1} , respectively. In the cases of liquid silicon at 1700 K, the values of A , B and C are 1.48×10^3 , -3.32×10^{-3} and -3.3 , respectively. These calculations were computed using the published physical property data [1,15]. The physical property values for the silica coating layer at high temperature (1473 and 1700 K) were assumed as the one tenth of those at room temperature which were supplied from its production company. Although this is a very extreme assumption, the latter term of Eq. (3) is small enough to ignore. Consequently, the presence of silica layer has no serious effect on the measurement and the thermal conductivity is derived from Eq. (1).

The measurements were carried out in the following manner. In the measurement of solid silicon, the experimental apparatus was located in the furnace. The temperature of the furnace was controlled within ± 1 K. In the measurement of liquid silicon and germanium, the sample was placed in a cylindrical alumina crucible (34 mm inner diameter and 150 mm high) and located in the furnace. To suppress steady convection during the measurement, the surface temperature of the sample was kept about 5 K higher than the bottom temperature. The probe was dipped vertically in the center of the sample, as shown in Fig. 1.

Generally, in experiments supplying the small current to the hot wire, the temperature rise of the hot wire is smaller. A smaller temperature rise might be better, resulting in less convection as well as delay of its onset. However, when the temperature rise is too small, it is very difficult to measure it accurately and the effect of heat capacity of the hot wire and silica layer cannot be ignored. On

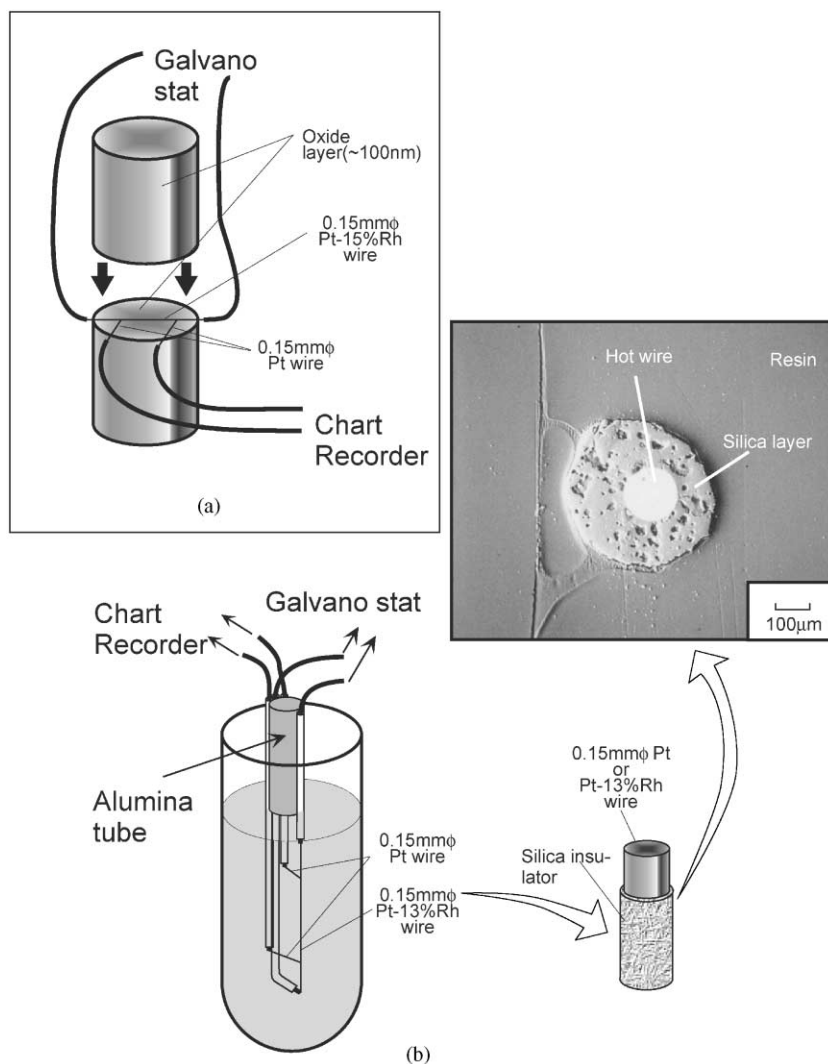


Fig. 1. Schematic diagrams of the experimental apparatuses for (a) solid silicon and (b) liquid silicon and germanium.

the other hand, the experiment using larger currents brings about convection at an earlier stage of the measurement. From the results of previous work [11], it was found that a current of 3.0 A provides suitable conditions, then the same condition has been employed as the supplied current in this study. The current was measured using the standard resistance. The voltage between the potential wires was monitored continuously using a chart recorder. These readings were then converted to resistance change of the hot wire and,

subsequently, to the temperature rise (ΔT) of the hot wire. The heat generation rate per unit length of the hot wire (Q) was calculated from the current and the resistance per unit length of the hot wire for each temperature. Consequently, the thermal conductivity of the sample was derived from Eq. (2), using values of Q and the slope of the linear relationship between ΔT and $\ln t$.

The samples of solid silicon were placed at room temperature in deoxidized argon in the furnace,

and then measurements were made at intervals of about 100 K during the heating cycle until 1673 K was attained, and were also made until about 673 K during the cooling cycle to confirm the reproducibility of the measurement. Measurements of liquid samples were carried out in homogeneous liquid state at temperatures between 1700 and 1724 K for liquid silicon and between 1273 and 1473 K at intervals of about 50 K for liquid germanium in deoxidized argon. At least two different probes were used for each sample and three runs were carried out at each temperature. After the experiments of both elements, the aluminum crucibles broke because of thermal expansion during the solidification but no accretion between alumina crucible and each sample was observed. This fact indicates that there was no reaction between crucible and each sample.

3. Results

3.1. Solid silicon

Fig. 2 shows a typical temperature rise (ΔT) as a function of logarithm of time obtained during measurements on solid silicon at 293 K. It can be

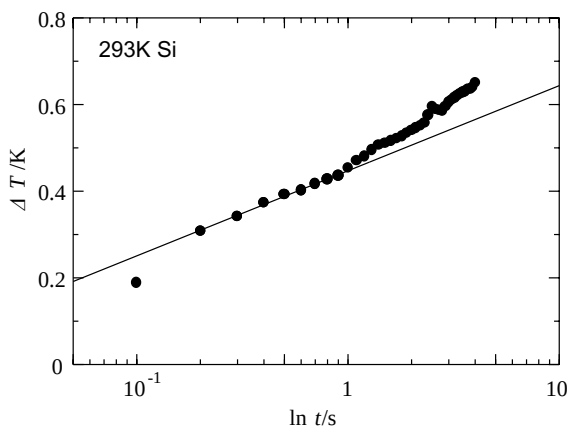


Fig. 2. Typical plot of a temperature rise as a function of logarithm of time for solid silicon at 293 K, where applied current is 3 A.

seen that there is a linear relation between ΔT and $\ln t$ for the time period, 0.2–0.9 s, and the temperature rise deviates from the linearity in the time period above 0.9 s. This deviation would be due to the effect of heat reflection at the edge of the sample. The effect of the heat reflection can be discussed using following empirical equation [11,13]:

$$t_d = 4.2 \times 10^5 \frac{r^2}{\lambda}, \quad (4)$$

where t_d is the time when deviation from the linearity starts, r the radius of the sample. It can be seen from this equation that the deviation starts earlier as thermal conductivity increases. In the present work, the radius of the sample was 15 mm and the thermal conductivity of solid silicon was measured as about 139.4 W/mK at 293 K. Using these values, a value of t_d is calculated as about 0.7 s and this value is in fairly good agreement with the time when the deviation starts. Accordingly, in the present work, the thermal conductivity values have been determined using the slope of the linear portion between ΔT and $\ln t$ in the time period 0.2–0.9 s.

Fig. 3 shows the temperature dependence of the thermal conductivities of solid silicon measured in the present work along with published data [3,6–8,14]. The closed circles refer to data recorded

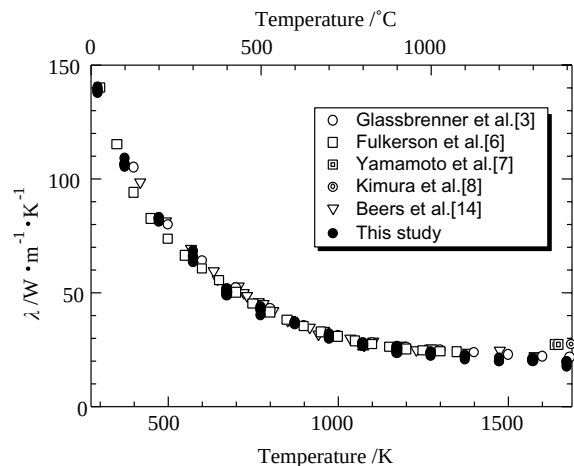


Fig. 3. Thermal conductivity of solid silicon compared with reported values.

in this study. The experimental uncertainty is within $\pm 5\%$, and the reproducibility of these measurements is very good. The thermal conductivities of solid silicon obtained in the present work were 139.4 W/mK at 293 K and 19.0 W/mK at 1673 K, and decrease faster than T^{-1} . This tendency is in good agreement with those reported by Glassbrenner et al. [3], Fulkerson et al. [6] and Beer et al. [14], where they measured on a polycrystalline silicon using the longitudinal heat flow method, on a cylindrical single crystal having (100) orientation using the radial heat flow method and on a single crystal using the diffusivity method, respectively. On the other hand, Yamamoto et al. [7] and Kimura et al. [8] measured thermal diffusivities by means of the laser flash method. Yamamoto et al. derived the thermal conductivity values from the data obtained. The values reported by Kimura et al. have been derived by the authors using the values of the heat capacity [15] and the density [8].

3.2. Liquid silicon and germanium

In the thermal conductivity measurement by the hot wire method, it is very important that there is no electrical leakage from the hot wire to the sample. To confirm this, the resistance of the hot wire between the potential wires was measured in air and in both liquids. In the former measurements, silica layer was not coated on the wire. Fig. 4 shows the resistances of hot wires with coating in these experiments, where the solid line represents the resistance of the hot wire with no coating. These results are in very good agreement with one another, which indicates that there was no electrical leakage from the hot wire to the melts.

Fig. 5 shows a typical temperature rise as a function of logarithm of time obtained during measurements on liquid silicon at 1710 K. It can be seen that there is a linear relation between ΔT and $\ln t$ in the time period from 0.2 to 1.0 s and that the temperature rise deviates from the linearity in the time period above 1.0 s. This deviation is due to the onset of convection [11]. The thermal conductivity is calculated from the slope of the linear

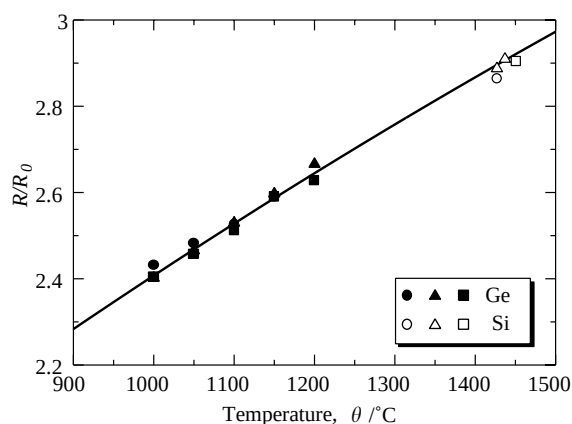


Fig. 4. Reduced electrical resistance (R/R_0) of the hot wire as various temperatures in the measurement of liquid silicon and germanium, where R and R_0 indicate electrical resistance and at each temperature and 0°C , respectively.

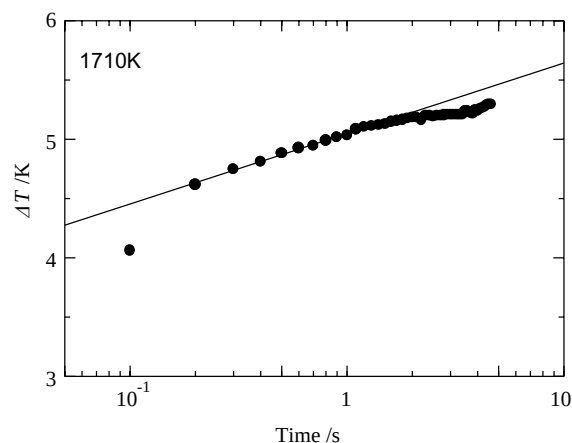


Fig. 5. Typical plot of a temperature rise as a function of logarithm of time for liquid silicon at 1710 K, where applied current is 3 A.

portion obtained between 0.2 and 1.0 s using Eq. (2).

Figs. 6 and 7 show the temperature dependencies of the thermal conductivities of liquid silicon and germanium, respectively, together with the reported values [7–10]. The closed circles refer to data recorded in the present work and the lines refer to values estimated from the Wiedemann–

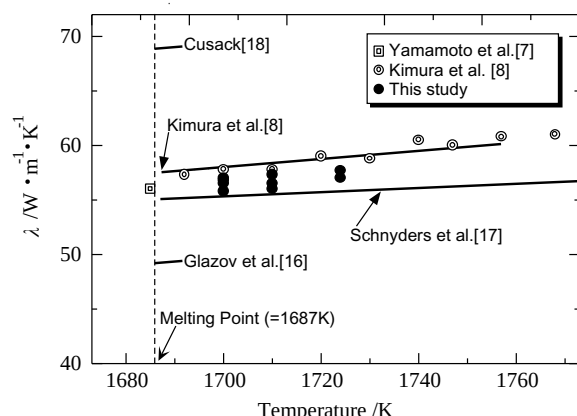


Fig. 6. Thermal conductivity of liquid silicon compared with reported values, where the values of lined are calculated from electrical conductivities based upon Wiedemann–Franz law.

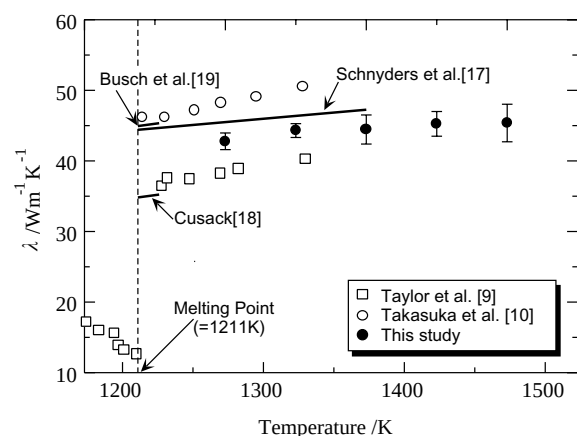


Fig. 7. Thermal conductivity of liquid germanium compared with reported values, where the values of lined are calculated from electrical conductivities based upon Wiedemann–Franz law.

Franz law using reported values of electrical conductivity [8,16–19]. The experimental uncertainty is within 1.2% and 5.9% in the measurements on liquid silicon and germanium, respectively, and the reproducibility of these measurements is very good. The thermal conductivities of liquid silicon recorded in the present work are 56.5, 56.6 and 57.4 W/mK at 1700, 1710 and 1724 K, respectively, which are slightly smaller

Table 1

Average values, standard deviations and scatters for thermal conductivities of solid silicon and liquid silicon and germanium at various temperature

Temperature (K)	Average value (W mK ⁻¹)	Standard deviation (W mK ⁻¹)	Scatter (%)
Solid silicon			
293	139.4	1.4	1.0
373	105.8	2.4	2.2
473	82.4	0.9	1.0
573	65.6	1.8	2.8
673	50.3	1.1	2.1
773	42.4	1.4	3.3
873	36.7	0.5	1.4
973	30.9	0.8	2.6
1073	27.4	0.7	2.5
1173	24.8	1.1	4.6
1273	23.7	0.5	1.9
1373	22.3	1.2	5.4
1473	20.1	0.8	3.8
1573	20.4	0.5	2.5
1673	19.0	0.9	4.7
Liquid silicon			
1700	56.5	0.5	1.0
1710	56.6	0.7	1.2
1724	57.4	0.5	0.8
Liquid germanium			
1273	42.8	1.2	2.8
1323	44.3	1.0	2.2
1373	44.5	2.1	4.6
1423	45.3	1.8	3.9
1473	45.4	2.7	5.9

than those obtained using the laser flash method by Yamamoto et al. [7] and Kimura et al. [8]. On the other hand, the thermal conductivities of liquid germanium recorded in the present work are 42.8 and 45.4 W/mK at 1273 and 1473 K, respectively, which lie between those by Taylor et al. [9] and Takasuka et al. [10]. The thermal conductivities for both materials tend to increase slightly with increasing temperature. Comparison between data recorded in the present work and those estimated from the Wiedemann–Franz law will be made in the following section. Table 1 gives a summary of average values, standard deviations and scatters of thermal conductivities for solid and liquid silicon and liquid germanium.

4. Discussion

4.1. Heat conduction mechanisms in solid silicon

In general, the thermal conductivity is expressed as follows:

$$\lambda = \lambda_l + \lambda_e, \quad (5)$$

where λ_l and λ_e are the lattice (phonon) thermal conductivity and the electronic thermal conductivity, respectively. The contribution from each mechanism will be discussed in the following section.

4.1.1. Phonon thermal conductivity

In solid silicon, phonon is the dominant heat carrier, since solid silicon is semiconducting and its electrical conductivity is rather small. It has been suggested by Glassbrenner et al. [3] that the phonon thermal conductivity is dominated by isotope, three-phonon and four-phonon scatterings and that the phonon thermal resistivity (W_l), i.e., the reciprocal of phonon thermal conductivity, can be described by

$$\lambda_l^{-1} = W_l = W_1 + BT + CT^2, \quad (6)$$

where the terms W_1 , BT and CT^2 denote the thermal resistivities due to isotope, three-phonon and four-phonon scatterings, respectively, and B and C are temperature independent constants. The term W_1 can be theoretically derived by using values of average volume per atom in the crystal, point impurity scattering parameter, Debye temperature and sound velocity in the crystal [3], and has a value of 2.92×10^{-5} mK/W in the present work. Therefore Eq. (6) can be reduced to the following equation:

$$\frac{\lambda_l^{-1} - 2.92 \times 10^{-5}}{T} = B + CT. \quad (7)$$

This equation means that the term in the left-hand side is linearly proportional to temperature. Fig. 8 shows a plot of the term in the left-hand side in Eq. (7) against temperature, where the value of λ_l^{-1} has been derived from thermal conductivity values measured in the present work. It can be seen from this figure that a good linear relationship applies at temperatures below ca. 1000 K. This

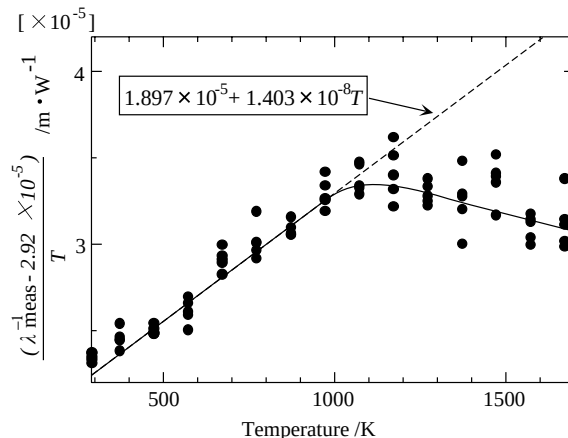


Fig. 8. The function $(\lambda_l^{-1} - W_1)$ to determine the extrapolated lattice thermal resistivity.

indicates that phonon is the dominant heat carrier in solid silicon at a temperature range below ca. 1000 K, and, as a consequence of this, the phonon thermal conductivity can be expressed by

$$\lambda_l = W_l^{-1} = \frac{1}{2.92 \times 10^{-5} + 1.897 \times 10^{-5} T + 1.403 \times 10^{-8} T^2} \quad (\text{W/mK}). \quad (8)$$

The deviation from the linearity above ca. 1000 K would be due to the other factor in Eq. (5), namely, electronic thermal conductivity.

4.1.2. Electronic thermal conductivity

Glassbrenner et al. [3] have also considered the effect of electronic thermal conductivity, and the same approach is taken in the present work too. The electronic thermal conductivity can be briefly explained as follows. In intrinsic semiconductors, there exist electrons and holes in the same concentration and these are produced by thermal excitation. Therefore, the contributions from electrons and holes to the thermal conductivity become increasingly more important at higher temperatures. The electronic thermal conductivity consists of two contributions. One is an electronic and hole polar contribution λ_{ep} , which is predicted

by the Wiedemann–Franz law. The other is an electron–hole bipolar contribution λ_{eb} , which is only characteristic of semiconductors as mentioned below: When there is a temperature gradient, the numbers of electrons and holes are greater in the hotter region. Consequently, both electrons and holes flow down from the hotter to the colder region. Some electrons and holes are recombined during the flowing process, releasing the energy in the colder region. The energy released by this recombination is nearly equal to the band gap energy. Accordingly, not only electrons and holes but also electron–hole pairs act as heat carriers in semiconductors at high temperatures, leading to Eq. (9)

$$\lambda_e = \lambda_{ep} + \lambda_{bp}. \quad (9)$$

These contributions are given by the following respective equations [20,21]:

$$\lambda_{ep} = 2 \left(\frac{k_B}{e} \right)^2 \sigma T, \quad (10)$$

$$\lambda_{bp} = \frac{b}{(1+b)^2} \left(\frac{E_g}{k_B T} + 4 \right)^2 \left(\frac{k_B}{e} \right)^2 \sigma T. \quad (11)$$

In Eqs. (10) and (11), e , σ , b and E_g represent the electronic charge, the total electrical conductivity, the mobility ratio and the band gap energy at T , respectively. Table 2 shows the calculated electronic thermal conductivities at various temperatures based upon literature values [6,22].

The experimental values of λ_e can be determined by subtracting the phonon conductivity expressed by Eq. (8) from the measured total thermal conductivity. Fig. 9 shows the value of λ_e

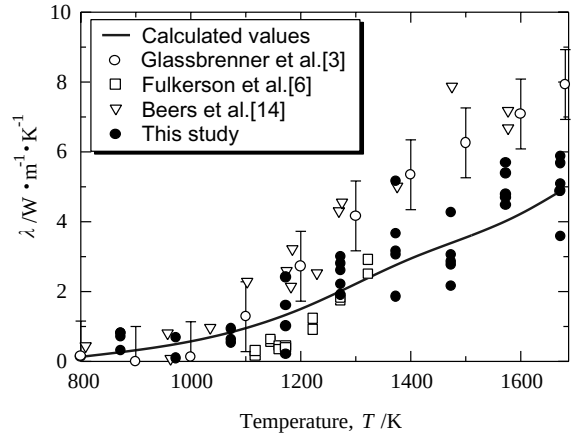


Fig. 9. Comparison of the experimental and theoretical thermal conductivity.

derived from measured thermal conductivities, together with values calculated from Eqs. (10) and (11). The values of λ_e derived from thermal conductivities obtained in the present work are in good agreement with the theoretical values. Consequently, it can be concluded that the thermal conductivity of solid silicon at temperatures above 1000 K includes contributions from electrons, holes and electron–hole pairs as well as phonons. Fig. 9 also includes reported data [3,6,14] measured by the steady-state method. The reported values are different from the theoretical values. The reason for this would be due to less accuracy in the measurements by the steady-state methods. It is very difficult to attain the steady state, especially at high temperatures and, in addition, it is difficult to confirm the establishment of the steady state.

Table 2
The calculated electronic thermal conductivities at various temperature [6,22]

T (K)	b	E_g (eV)	σ ($\Omega^{-1} \text{ m}^{-1}$)	λ_{ep} ($\text{W m}^{-1} \text{ K}^{-1}$)	λ_{bp} ($\text{W m}^{-1} \text{ K}^{-1}$)	λ_e ($\text{W m}^{-1} \text{ K}^{-1}$)
800	0.31	0.92	3.91×10^2	0.005	0.13	0.13
1000	0.31	0.85	2.1×10^3	0.031	0.54	0.57
1200	0.31	0.78	6.46×10^3	0.110	1.39	1.50
1400	0.31	0.71	1.44×10^4	0.300	2.65	2.95
1600	0.31	0.55	2.63×10^4	0.630	3.60	4.23
1671	0.31	0.52	3.14×10^4	0.780	4.08	4.86

4.2. Heat conduction mechanism in liquid silicon and germanium

In general, the thermal conductivities of liquid metals are smaller than those of solid metals, since the mean free path of electrons as heat carriers is shorter in liquid state. However, from Figs. 6 and 7, it has been found that the thermal conductivities of both liquid silicon and germanium are greater than those of solid silicon measured just below the melting point. A similar phenomenon has been observed in electrical conductivities of liquid silicon measured by Sasaki et al. [23], who has suggested that electrical conductivities of liquid silicon are close to those of typical liquid metals such as mercury.

It has been suggested by several researchers that the Wiedemann–Franz law applies to some liquid metals. This law has been derived from the free electron model on the assumption that the relaxation times are identical for electronic and thermal processes. The Wiedemann–Franz law states that the thermal conductivity is proportional to the product of the electrical conductivity (σ) and the absolute temperature as follows:

$$\lambda = L\sigma T, \quad (12)$$

where L is the Lorenz number ($=2.45 \times 10^{-8} \text{ W}\Omega/\text{K}^2$).

With respect to liquid silicon, the thermal conductivity values recorded in the present work are close to those derived from the Wiedemann–Franz law by Kimura et al. [8] and Schnyders et al. [17], as shown in Fig. 6. On the contrary, thermal conductivity values calculated using electrical conductivity data reported by Glazov et al. [16] or Cusack [18] are very different from the others. However, the temperature dependence seems to be almost the same. In liquid germanium, the thermal conductivity values in the present work are close to values derived from the Wiedemann–Franz law by Schnyders et al. [17] and Busch et al. [19], but are greater than values derived by Cusack [18]. However, there is good agreement with one another in the temperature dependence, which is same as the case of liquid silicon.

One of the reason for this would be the difference of the purification of the sample due to impurities from the crucible made from high density graphite [17], boron nitride [8] and so on. In general, the effects of impurities to the electrical conduction and thermal conduction are same [24]. Therefore, when the electrical conductivity and thermal conductivity are measured using same samples, the thermal conductivity value calculated from the measured electrical conductivity would be close to the measured thermal conductivity within experimental error, if the Wiedemann–Franz law can be applied. In practice, Kimura et al. [8] have measured electrical conductivity and thermal diffusivity using the same silicon sample, and the thermal conductivity value calculated from the measured electrical conductivity is in good agreement with the thermal conductivity calculated from measured thermal diffusivity as shown in Fig. 6. This fact is very interesting and would be one of the effective evidences for applicability of Wiedemann–Franz law to the liquid silicon.

The other possible reason would be the reasonability of the Lorenz number. In the above estimation of thermal conductivity, a value of $2.45 \times 10^{-8} \text{ W}\Omega/\text{K}^2$ has been used as the Lorenz number, where this value has been derived based upon the free electron model. But in actual metals, this value is not always suitable even in solid state because the electron is not perfectly free but nearly free. On the nearly free electron model, the Lorenz number would have a value different from $2.45 \times 10^{-8} \text{ W}\Omega/\text{K}^2$. This indicates that the absolute value of the measured thermal conductivity is different from that calculated from the electrical conductivity on the free electron model but both temperature dependence is same. In practice, such phenomena are frequently found in the actual metals [1]. From this viewpoint, it can be concluded for liquid silicon and germanium that the electrons behave as nearly free electrons and heat is predominantly transferred by nearly free electrons similar to typical metals. This conclusion leads to the consideration that the electronic structure of liquid silicon is similar to that of liquid metal.

Busch et al. [19] have also concluded that Hall coefficients of liquid germanium and metals are close to values derived from the free electron model, which supports the consideration mentioned above.

5. Conclusions

The thermal conductivities of silicon and germanium have been measured in solid and liquid states using the non-stationary hot wire method with a silica-coated probe.

- The thermal conductivities of solid silicon are 139.4 ± 1.4 W/mK at 293 K and decrease faster than the reciprocal of temperature (T^{-1}), being 19.0 ± 0.9 W/mK at 1673 K.
- In the thermal conductivity of solid silicon, phonon is the primary heat carrier at temperatures below ca. 1000 K, whereas above ca. 1000 K electronic mechanisms due to electrons, holes and electron–hole pairs are also operative.
- The thermal conductivity of liquid silicon is 56.5 ± 0.5 W/mK around 1700 K and slightly increases with increasing temperature.
- The thermal conductivities of liquid germanium are 42.8 ± 1.2 W/mK at 1273 K and 45.4 ± 2.7 W/mK at 1473 K.
- The thermal conductivity values for liquid silicon and germanium are close to those calculated from electrical conductivity values based upon Wiedemann–Franz law, respectively, which indicates that heat is predominantly transferred by nearly free electrons in both liquid elements.

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